

Report conversation

I am attempting to summarize an academic paper with the following abstract:

abstract:

To search for an unknown sign in a sign language dictionary, users typically indicate parameters of the query, e.g., hand shape and signing location. Recent advances in sign language recognition enable video-based sign language dictionary search. In such a system, users can record an unknown sign and retrieve a list of signs that look similar, preferably including the queried sign as one of the top results. We have realized such a system by interpreting it as a dense vector search task. First, we learn a mapping (embedding) from sign videos to a vector space. The dictionary can then be searched by looking for the vectors in this space that are closest to the vector corresponding to the query. We present a proof of concept on a subset of the Flemish Sign Language dictionary. Further research is required to scale up our method to the large vocabularies of entire dictionaries.

Can you make the following academic paper summary a bit more concise, while still being accurate to the abstract?

Summary:

@costerQueryingSignLanguage2023 describe a proof of concept system which would allow users of an online dictionary of signs to search by signing, a more user-friendly method.

That is, by recording and uploading a video of themselves signing, one can query the dictionary that way.

They do this by essentially by finding the video with the most similar embeddings.

They start by first pretraining a Sign Language Recognition (Pose sequence to Gloss) model, thus training their transformer encoder to embed sign inputs.

<!-- Specifically a modified PoseFormer, according to the Repo. In the paper, Coster et al cite Sign Language Recognition with Transformer Networks (2020).

<https://aclanthology.org/2020.lrec-1.737/> -->

Once the encoder is trained, then they can use it to generate embeddings for all the signs in the dictionary. When a user input query video arrives they can then compare the embeddings of the input to those of dictionary entries using Euclidean distance.

Tests on a Flemish Sign Language dictionary show that it can successfully retrieve a limited vocabulary of signs, including some it was not trained on.

[A prototype repository is provided]

(<https://github.com/m-decoester/VGT-SL-Dictionary>), including scripts to generate a database for a dictionary as well as documentation for installation or training of new models.

@costerQueryingSignLanguage2023 present a proof of concept for a sign language dictionary that allows users to search by signing. Users can record a video of themselves signing and upload it to query the dictionary. This system works by finding the video with the most similar embeddings. The process begins with pretraining a Sign Language Recognition model to embed sign inputs. Once the encoder is trained, it generates embeddings for all dictionary signs. When a user submits a query video, the system compares the input embeddings with those of the dictionary entries using Euclidean distance. Tests on a Flemish Sign Language dictionary show that the system can successfully retrieve a limited vocabulary of signs, including some not in the training set. A prototype repository is available, including scripts for database generation and documentation for model installation or training.